CHALLENGE

01

Seconds Ahead

Every crash on an Indian road gives a warning first. Build the thing that catches it.

POWERED BY

PCB Cupid

Hardware kit partner · kit available on site, optional

Hacktivate 2026 · 24-hour Hardware + AI Hackathon · Theme: Predictive Systems
14 Oct 2026, 11:00 AM to 15 Oct 2026, 11:00 AM · Pravrutti 2.0 (National Tech Fest)

Goals and Motivation

Meet Imran (one example of many)

Imran is 38 and drives an autorickshaw out of Peenya. Every morning at 7:40 he picks up two schoolchildren from a narrow cross road and turns onto the main road. A lorry is almost always parked at the corner. From the driver's seat, he can see its tyres and nothing else, so he does what every driver in India does: honks, noses out, and hopes.

One Tuesday a scooter is already there. The rider brakes, skids and goes down a metre from the auto's front wheel. Nobody is badly hurt, but the children are crying and Imran is shaking, because he knows it wasn't care that saved them. It was luck. And none of it was unpredictable: the scooter had been coming, fast and loud, for five seconds before he moved.

Imran's corner is just one example, not the challenge. The same story plays out a thousand different ways every day. A delivery rider rear-ended at a signal by a lorry that never slowed. A schoolgirl crossing a road where bikes don't stop. A scooter that drops into an unmarked pothole after dark. A bus overtaking a truck on a two-lane highway into a vehicle it can't see. An elder stepping off a bus into a squeezing two-wheeler. A rider going the wrong way down a one-way stretch to save two minutes. Each of these is a different problem. What they share is that the danger was building for seconds, sometimes minutes or days, and nobody was told in time.

This is not a problem of missing technology. Speed, distance, sound, light, motion, location and behaviour are all measurable, and cheap boards can learn patterns. What is missing is **products built for ordinary people on Indian roads that see danger coming and warn them, or act, before it happens.** The expensive version exists in premium cars. The people who die most on our roads walk, ride two-wheelers, drive autos and take the bus, and they have almost none of it.

The Scale

India recorded **4,80,583 road crashes and 1,72,890 deaths in 2023**, about 20 deaths every hour (MoRTH, Road Accidents in India 2023). Use these numbers as evidence of how big the problem space is, and as a starting point for your own research, not as a list to pick from:

What the data shows (India, 2023)	Deaths	Source (MoRTH RAI 2023)
Two-wheeler riders killed (44.8% of all deaths)	77,539	Table 4.4
Pedestrians killed (20.4%, up 7.3% in a year)	35,221	Table 4.4
Hit from behind, the biggest single collision type	36,804	Table 1.5
Crashes at junctions (19.9%); T-junctions alone 12,311	34,354	Table 3.6
Crashes on curved roads	22,263	Table 3.5
Riders and pillions killed without helmets	54,568	Table 3.3
Over-speeding (68.1% of deaths)	1,17,682	Table 3.1
Driving on the wrong side / lane indiscipline	9,432	Table 3.1
People under 18 killed	9,489	Table 4.2
Crashes on pothole roads	2,161	Table 3.5

The deadliest window is **6 PM to 9 PM, with 20.8% of all crashes** (Table 7.3). Worldwide, road crashes kill about **1.19 million people a year** and are the leading killer of children and young people aged 5 to 29; more than half of those killed are pedestrians, cyclists and motorcyclists (WHO Global Status Report on Road Safety 2023).

You are on these roads too. Somewhere on your own commute there is a moment like Imran's corner. This challenge asks you to find it, and to put yourself, and everyone like you, seconds ahead of the danger.

Three Failures. One Challenge.

Everyday road safety in India fails ordinary people in three ways at once:

WARNINGS ARRIVE AT IMPACT	Horns, brake lights, mirrors and speed breakers all work in the last second, or after it. By the time a rider sees the brake light, a walker hears the horn or a driver spots the pothole, the moment to act has already passed. Almost nothing on the road says what is about to happen.
SAFETY TECH IS BUILT FOR THE FEW	Collision warnings, blind-spot detection and driver-assistance systems mostly ship in expensive cars. The people who die most often — pedestrians, two-wheeler riders, auto drivers, bus passengers, children — rarely have any of it, and the roads they use most get the least attention.
THE ROAD IS DIFFERENT EVERY DAY	Indian traffic is mixed and changes by the hour: school rush, delivery peaks, cattle, wrong-side riders, rain, darkness, a new pothole. Static signs and fixed rules can't know this corner, this hour or this rider. Nothing learns the road and the people on it.

Your Challenge

Build any product that predicts a danger on India's everyday roads before it happens — for a road user you are yourself, and test it on yourself.

Pick a road user your team really is, or can genuinely become this week: you ride a scooter or bike to college, ride pillion, walk to the bus stop, take autos, cabs or buses, cycle, or are learning to drive. **Your team lead is your primary user.** Watch your own commute, find the danger or near-miss that actually happens to you, and build the product you would want on your own way home. It can be a wearable, a helmet or bag add-on, something on your scooter or cycle, a gadget for your bus or auto ride, a roadside unit, an app working with hardware, or something nobody has built yet.

A forecast, not an alarm

Your product must see the danger coming: how soon, how likely, or how a pattern is drifting towards trouble. "Something is close" or "you crashed" is a reaction. "That bike won't stop in time", "this stretch gets dangerous in 10 minutes" or "your riding has been getting riskier all week" is a prediction.

Prediction is only half the job. The forecast has to reach the right person, in a form they can use in the moment, early enough to change what happens.

The Rules

- **Any components, any form.** Use any hardware, sensors, boards, phones, cameras or software you like. PCB Cupid kit parts are available on site if you want them; you can also bring your own.
- **Real hardware in the loop.** This is a hardware + AI hackathon: something physical must sense the world or act in it. A phone or app can be part of your system.
- **A real prediction.** Time-to-event, an early-rising risk score or a learned pattern, as defined in The Prediction Requirement below. Crash detection after the fact and simple threshold alarms don't count on their own.
- **Be the user.** Choose a road user your team really is or can become. The team lead is the primary user, observes their own commute, and tests the product on themselves (see Be the User below).
- **Test safely, always.** Self-testing happens walking, on a parked or stationary vehicle, in a quiet campus lane or on a mock setup. Never test in live traffic in a way that puts anyone at risk.

Building Blocks

Most predictive safety products are made of these four pieces. They describe the thinking, not the parts: use whatever components fit your idea.

01 Be the User

Before you build: watch your own road. The team lead (and anyone else in the team who shares the commute) observes their own daily journey — to college, to the bus stop, to the hackathon itself. Log every near-miss, scare or annoyance: a photo of the spot, a quick note of what happened and when, a short voice memo recorded right after (stop safely first). Two or three honest entries beat a long imagined list.

During the 24 hours: test it on yourself. Wear it, carry it or fit it, and try it in safe ways: walking, on a parked or stationary vehicle, in a quiet campus lane, or on a mock setup on the table. Every time, note what you tried, what the device predicted, what actually happened and what you changed.

At the end: be honest. Would you actually use this every day? What would make you switch it off? What would make you buy it?

Required: Bring a short **commute log** (your real near-misses or annoyances, with at least one photo, note or voice memo) and a mini **test log** of your own self-tests (see the example below), plus one sentence answering "Would I actually use this daily?"

What we tried	What it predicted	What actually happened	What we changed
Walked past a parked scooter that started to pull out (mock setup in the lab)	"Vehicle moving, about 2 s"	Warning came, but too late to step aside	Raised the sampling rate; warning now about 3 s ahead
Lead wore the helmet unit on a stationary scooter while a teammate rolled a cycle past	Warned on every pass, even slow ones	Annoying; lead said they would switch it off	Added a speed check; slow passes now stay silent

Example test log format only. Yours will look different.

02 Seeing It Coming

Find the signals that change before the danger arrives: a closing speed, a sound getting louder, light changing, a vehicle's motion, a time of day, your own riding or walking habits, a spot's history. Capture them with whatever sensors suit your product, and log real data from your own commute or self-tests early, because your forecast is only as good as what it has seen.

03 The Forecast

Turn the signals into a statement about the future: "impact risk rising, about 3 seconds", "this junction on your route is busiest and most dangerous between 8:30 and 9", "your braking has been getting later all week". Show how far ahead and how sure you are. Simple, honest models often beat impressive-sounding ones.

Required: A real prediction, as defined in The Prediction Requirement below.

04 Acting in Time

Get the forecast to you in a way that fits your moment: a tap on the head, a glow at the edge of your vision, a sound you can hear through traffic, a light that warns the driver behind you, a voice in your ear. If it annoyed you during self-testing, it will annoy everyone. A warning you learn to ignore, or one that cries wolf, is worse than none.

Stretch Goals

- **Learns the road.** Your product adapts to a specific route, junction, rider or time of day, and gets better with use.
- **Warns both sides.** The person in danger and the person causing it both find out in time.

- **Cheap enough to matter.** A price that a student, a rider, an auto driver, a school or a small fleet would actually pay.
- **Day, night and rain.** It still works when the road is at its worst.
- **Voice.** Speak the forecast with ElevenLabs (credits for every team) — see Where Voice Fits below.

Where Voice Fits

Riding in traffic is a hard place to listen to long sentences, so think about **who** hears the voice, **when** and **how much**. Voice is optional on this challenge, but it fits the road better than you might expect, and every team gets **ElevenLabs credits** to try it:

- **Passengers** — on a bus or in an auto, people are free to listen: "Hold on, sharp brake ahead."
- **Walkers** — a roadside unit at a crossing or school gate that says "Wait, bike coming", or "Clear now, cross".
- **One word in the helmet** — a single calm word through a helmet earpiece or bone-conduction speaker ("Left!", "Slow"), not a conversation.
- **Learner drivers** — a co-pilot voice that predicts the tricky moment on the practice route and talks the learner through it.
- **Before the ride** — a pre-ride briefing: "Rain on your route, the underpass floods by 6; leave ten minutes early or take the flyover."
- **After the ride** — a coach that reviews your near-misses by voice once you've stopped: "You braked late at the same junction three days in a row."
- **Riders sharing alerts** — short spoken warnings passed between riders about what's ahead: "Fresh pothole after the signal."

Keep it safe: short, calm, non-distracting prompts at the right moment. Never ask a rider or driver to look at a screen or hold a conversation while moving. Voice through any speaker, earpiece, phone or laptop is fine, in any language.

Sparks, Not Assignments

These sparks point in deliberately different directions, to show how wide (and how fun) this challenge can be. **Please don't copy them.** The best projects will come from something that actually happens on your own commute. Pick the road user you are and a moment:

Which road user are you?	Spark moments
Scooter or bike rider	The morning rush to college
Pillion rider	The evening ride home, 6 to 9 PM
Walker to college or the bus stop	Night on an unlit road
Auto, cab or bus passenger	Getting on or off a bus or auto
Cyclist	The first rain of the season
Learner driver	Anything from your own commute log

- **A helmet that taps before the pothole** — it learns your daily route and gives a little tap on the head (left, right or both) a few seconds before the bump you always forget.
- **A jacket or backpack that glows at the car behind** — it predicts that the vehicle behind you isn't slowing and lights up brighter and faster until it does.
- **A "should I cross now?" buddy** — a clip-on or roadside unit for walkers that reads the traffic, predicts the next safe gap and says "wait, bike coming" or "go now" in a voice you trust.
- **A ride coach with a personality** — a scooter or cycle add-on that learns your style, then, once you've parked, reviews the day's near-misses in a voice that teases you, kindly, when your braking or speed starts drifting towards trouble.
- **A bus or auto co-pilot** — something passengers carry that predicts the hard brake or the sharp turn and calmly says "hold on" just before it.

- **Riders warning riders** — gadgets that share what's ahead (a fresh pothole, a stalled bus, a wrong-side rider) as a short spoken alert, so the next person on your route hears it before they get there.
- **Something nobody has built yet** — the most original idea in the room.

The Prediction Requirement

This is a Predictive Systems hackathon. A product that reacts after the danger has arrived is an alarm, not a forecast.

Your device must make a **real prediction**. That means it forecasts something that has not happened yet, in one of these forms:

- **Time-to-event** — "about 40 seconds until this happens", updated as the situation changes.
- **An early-rising risk score** — a likelihood that climbs as precursors build up, well before the event itself.
- **A learned pattern** — the device learns what normal looks like for this person, place or machine, and flags the drift away from it before it becomes a problem.

Threshold alarms do not count. "Beep when the distance drops below 50 cm" or "buzz when the temperature crosses 40°C" is a reaction, not a prediction. A real forecast says what is coming and how soon, before any threshold is crossed.

Examples of what a real prediction looks like for this challenge:

- A rider-worn device that estimates seconds-to-impact from a vehicle closing in, and stays calm when the same vehicle is close but slowing.
- A school-gate unit that learns the traffic at that spot, predicts the next safe gap and tells children out loud, "Wait… now you can cross", updating as the traffic changes.
- A vehicle add-on that learns a driver's normal and flags the drift towards risky driving days before something happens.

Required: Show what your product predicts, how far ahead and how sure it is, and show it working live in your demo.

Be Your Own First User

This challenge is built for and by real road users: you. Your team lead is the primary user, and the product should solve something that really happens on their commute. You are welcome to also talk to someone who shares that road (a friend who rides the same route, the auto driver who takes you to college), but your own experience and self-testing come first.

Required: Bring your **commute log** (real near-misses or annoyances, with at least one photo, note or voice memo), your **test log** from safe self-tests during the 24 hours, **one design change** your own use caused, and an honest answer to **"Would I actually use this daily?"**

The Demo Requirement

Hacktivate is judged on what works in front of the judges. There are no slides. Your demo is the device, your KiCad design and the person you built it for.

- **Live, no slides.** You present the working hardware on the table. A phone or laptop may show your sensor readings, forecast plot or serial monitor, but not a pitch deck.
- **KiCad circuit.** Show the schematic (and PCB layout, if you got that far) for the circuit you actually built. It should match the hardware on the table.
- **Working hardware.** A simulation, video or Wokwi model does not count as the demo.
- **Show it working.** Run a live demo of your product doing its job: sensing, making its prediction and acting on it.
- **Show your logs.** Your commute log and test log are part of the demo; keep them on a phone, laptop or paper on the table.

- **Bring the road to the table, your way.** A table model with toy vehicles, a recording from your own self-test replayed through your device, a parked vehicle outside, or something else entirely. You choose the setup.

Required: All of the above, in a demo of about five minutes, followed by questions.

Resources

Hardware

- **Any hardware is allowed.** Any microcontroller, SBC, phone, sensor, camera or actuator you can run safely.
- **PCB Cupid kit (optional).** Kit parts are available on site if you want them: ESP32-H2 and ESP32-C6 boards, ultrasonic, IR, hall, LDR and microphone sensors, MPU6050 accelerometer/gyro, DHT11, RFID, rotary encoders, a 16x2 LCD, 7-segment displays, buzzers, LEDs and RGB LEDs, and prototyping supplies. Quantities are limited, so ask at the help desk.
- **Bring your own.** Anything your idea needs beyond that.
- **Provided by organisers:** ElevenLabs credits for every team.

Tools and Methods

- **KiCad** for the schematic and PCB layout.
- **Simple, explainable methods first** — time-to-collision from distance and speed, rate of change, moving averages, Kalman filters, regression over a sliding window, anomaly scores against a learned baseline.
- **Tiny ML or on-device vision** — TensorFlow Lite for Microcontrollers, Edge Impulse or anything else that fits.
- **Open data** — MoRTH's Road Accidents in India report and city crash data can help you find where and when your problem is worst.
- **ElevenLabs** credits for every team, if your product speaks.

What Makes a Strong Submission

Strong Submissions...	Weak Submissions...
— Be the user. The lead is a real road user of this kind, and the team can show the near-misses from their own commute that started the idea.	— Build for an imagined user and invent their needs the night before.
— Test it on yourselves, safely. A test log shows what you tried, what the device predicted, what happened and what you changed.	— Never try it outside the code editor, or only test it once, right before judging.
— Iterate from your own use. At least one real design change came from wearing, carrying or riding with the product.	— Keep the first version unchanged because "it works on the bench".
— Predict before it happens. The forecast leads the danger, states its lead time and confidence, and works live in the demo.	— Detect a crash after it happens, or beep at a fixed threshold, and call it predictive.
— Be original and honest. A fresh product or angle, plus a straight answer to "Would I actually use this daily?", including what still annoys you.	— Copy one of the sparks, or claim it is perfect and everyone will love it.

— Demo it live. Working hardware on the table and a forecast that responds.	— Show slides, a video or a simulation instead of a working product.

Evaluation Criteria

Every challenge is judged out of 20 marks. Creativity breaks ties.

Criterion	Marks	Weight	What judges look for
KiCad design	5	25%	A clean schematic for the hardware you built, matching the device on the table; sensible power and signal design. Any components are fine.
Hardware concept and working (incl. prediction quality)	7	35%	Is the problem real, found on your own commute? Is the prediction real, early and honest about confidence? Is the product original, does it act on the forecast in time, and did it improve through your own testing?
User understanding	3	15%	Be the User evidence: a real commute log, a safe self-test log, design changes driven by your own use, and an honest "would I use this daily?" reflection.
Live demo	5	25%	Live, no slides; clear in about five minutes; the product works in front of the judges.

Ground Rules

- **Test safely.** Self-test by walking, on a parked or stationary vehicle, in a quiet campus lane or on a mock setup. Never test in live traffic in a way that puts anyone at risk, and never record voice memos or notes while riding or driving. Anything that acts on a real vehicle (brakes, throttle, steering) must be demonstrated on a model.
- **Assistive and early-warning aids only.** No medical diagnosis claims. "Your posture is drifting" is fine; "you have a spinal condition" is not.

Why This Matters

India loses more people on its roads than almost any country on Earth, and most of them are not in cars. They are walking to work, riding to a delivery, driving an auto, taking children to school. For them the road is an everyday risk taken with nothing but eyes, ears and a horn, and every crash arrives as a surprise even though the warning was there seconds, minutes or days before.

Sensors are cheap, boards can learn, and every one of you uses these roads. If you can build a product that sees the danger first and gets the warning to the right person in time, for people who have never had that kind of foresight, you won't just win a hackathon. You'll have built something that could keep someone you know alive.

Appendix — Contact & Additional Resources

Challenge contact: Ganesh S, Chair, IEEE SB, Ramaiah University of Applied Sciences · voxmastery@ambugo.tech · +91 79753 24769

Web: pravrutti.tech · msruas.ac.in

- Kit help desk: on site at Pravrutti 2.0 for PCB Cupid kit parts (optional for this challenge).
- Road safety data: Ministry of Road Transport and Highways (MoRTH), Road Accidents in India 2023 — Tables 1.5, 3.1, 3.3, 3.5, 3.6, 4.2, 4.4 and 7.3. morth.gov.in
- Global context: WHO Global Status Report on Road Safety 2023 (who.int).
- KiCad documentation: docs.kicad.org · ElevenLabs: elevenlabs.io/docs